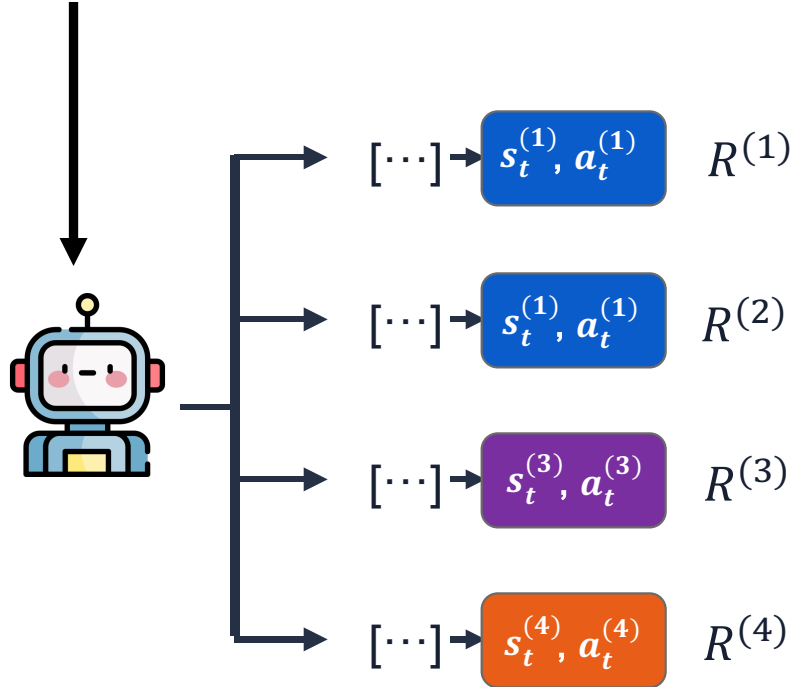


Rollout Collection (G=4)

heat some egg and put it in countertop.



all $(s_t^{(i)}, a_t^{(i)}, R_t^{(i)})$ from one group

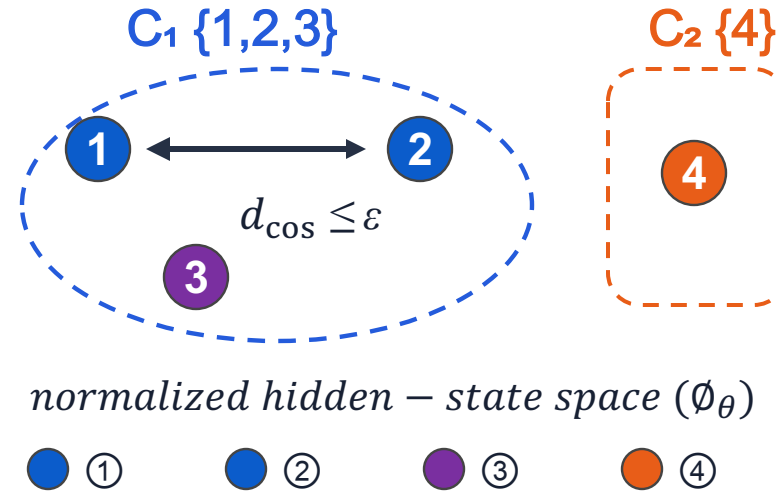
step
records

BiGPO — Actor-Hidden Clustering

Extract hidden state $f_\theta(s_t)$ at layer -8

Normalize: $\phi_\theta(s) = f_\theta(s) / \|f_\theta(s)\|_2$

Greedy cosine cluster: join if $d_{\cos} \leq \epsilon$



clusters

PACE — Action-Conditioned Baseline

Take cluster $C \in \mathcal{C}^{\text{BiGPO}}$, e.g. $C_1 = \{1, 2, 3\}$

Action key $x(a_i)$: first $-N-$ token hash or $\langle \text{action} \rangle$ tag

cluster C_1 (bisim – equivalent states)

$C_1^{(i)} = \{1, 2\}$

same action a_1

1

$R^{(1)}$

2

$R^{(2)}$

$C_1 \neq^{(i)} = \{3\}$

diff. action a_2

3

$R^{(3)}$

$$\hat{Q}_{(s, a_i)} = \text{mean}\{R^{(j)} : j \in C^{(i)}\}$$

$$\hat{V}_{(s)} = \text{mean}\{R^{(j)} : j \in C\}$$

$$\hat{A}_i^s = \hat{Q}_{(s, a_i)} - \hat{V}_{(s)}$$